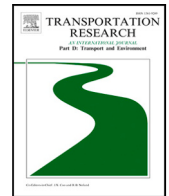




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Data-driven analysis of strategic–operational interfaces in freight electrification under deep uncertainty

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ABSTRACT

Battery electric trucks (BETs) offer environmental benefits but have been challenged by technical and economic viability compared to internal combustion engine trucks (ICETs). Meanwhile, fleet owner-operators have difficulties making strategic decisions of freight electrification based on Total Cost-of-Ownership (TCO) due to inaccurate representations of fleet operations, high uncertainty, and lack of generalizability. We therefore develop a data-driven analytical framework for BET and ICET fleets evaluating over 100,000 scenarios to measure the effects of strategic decisions, operational optimization, and deep uncertainty on TCO and competitiveness. We find that electrification becomes more efficient at scale, enabling optimization of routing, charging and scheduling to better mitigate operational constraints, thus improving service level and utilization. Partial electrification in large networks shows to be a robust pathway towards uncertainties such as battery prices and future battery technology. The study results provide guidelines for fleet operators realizing electrification and thereby their sustainability goals in operations.

1. Introduction

The electrification of road transport has emerged as a key priority for policymakers and industry leaders globally, driven by the urgent need to reduce greenhouse gas emissions and mitigate climate change (Rogstadius et al., 2024). The introduction of battery electric trucks (BETs) offers environmental advantages over conventional internal combustion engine trucks (ICETs) with lower carbon emissions and better energy efficiency (Juan et al., 2016; Macharis et al., 2013) and has been identified as the key path towards decarbonizing freight transport (Rogstadius et al., 2024).

However, cost-competitiveness of BETs relative to ICETs is identified as the primary barrier to the adoption of electric freight (Naumanen et al., 2019; Macharis et al., 2013), and has received substantial research attention. Despite this, logistics service providers have stated that commonly used models for cost assessment, such as total cost of ownership (TCO), are unsuitable for supporting strategic decisions due to a lack of consideration of the operational implications of freight electrification (Schiffer et al., 2021), making fleet investment decisions highly difficult.

Comparing BET costs with ICETs poses several challenges. On the level of strategic decisions, unpredictable market dynamics such as battery prices and technological developments in relation to processes such as selection of system and sub-system features, e.g. battery capacity, make it difficult to determine the number of vehicles to electrify and the associated battery capacity. Furthermore, the operational decisions involved in operating an electric fleet are also sensitive to market dynamics and conditions, such as

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energy prices and demand, which are inherently unpredictable. Additional uncertainty is introduced with unpredictable and complex interactions between these strategic and operational decisions. Such high degrees of uncertainty, complexity, and non-linear behavior require more robust methods for decision-making. From an application aspect, previous studies on freight electrification have been made at the vehicle or aggregate level, but are unable to provide guidelines for decision making that present a full picture of adopting electrification, nor evaluating the interfaces between strategic and operational decisions at the fleet level.

There is therefore a significant research gap regarding these interfaces; how strategic decisions regarding factors such as battery capacity, charging infrastructure, and network structure interact with operational planning of routes, vehicle selection, and charging, and how such decisions can be made to account for deep uncertainty, which is highly relevant to fleet operators and electrification of road freight globally. In this study, we tackle the above challenges by introducing a decision making under deep uncertainty (DMDU) framework which provides data-driven analytical tools for high degrees of uncertainty and non-linearity, and can potentially be used to gain insights into the most significant factors that affect the TCO of BET fleets and their cost-competitiveness relative to ICET fleets, including optimization of charging, route selection, and scheduling.

By using industry data and considering a wide range of scenario parameters, including market and technology dynamics and operational features, quasi-Monte Carlo methods such as Sobol sampling and global variance decomposition provide a systematic analysis and present a pathway for the decarbonization of road freight. The scope of this study is distribution networks of up to 20 sites in a radius of 20 to 250 km from the distribution center, considering fleets of up to 20 battery-electric or diesel 40 tonne semi-trailers.

2. Literature review

2.1. Barriers to electrification

Barriers to the adoption of BETs include technical, financial, and operational factors, particularly the significant upfront capital costs, the limited availability of charging infrastructure, and concerns regarding BETs' range and payload capacity (Lebeau et al., 2016; Taefi et al., 2016; Rogstadius et al., 2024; Schiffer et al., 2021; Bae et al., 2022; Cunanan et al., 2021) and lack of operational data outside of pilot projects (Anderhofstadt and Spinler, 2019).

Electrified transport systems contain complex dynamics and interactions between the properties of the vehicles, batteries, chargers, shipments, charging infrastructure, vehicle routing, and electricity prices (Rogstadius, 2022). Complex interactions and dynamics tend to create systems that are highly unpredictable and uncertain, thus making effective and robust decision-making difficult (Kwakkel and Haasnoot, 2019). Furthermore, this complicates evaluating how different parameters influence the total cost of ownership (TCO) and cost-competitiveness of freight electrification, thereby inhibiting strategic decisions such as investment in BETs and associated charging infrastructures. The effect of freight electrification on system-level logistics performance, particularly how planning is affected by operational, has thus been identified as a significant research gap in the literature (Gillström et al., 2024).

2.2. TCO analyses of freight electrification

Transportation buyers are highly sensitive to cost-competition among logistics service providers (Winston, 1981), which is affected by capital-intensive fleet acquisition costs (Anderhofstadt and Spinler, 2019). TCO models have a broad view of internal and external costs for evaluating strategic competitive advantages (Van Weele, 2002; Ellram and Siferd, 1998) and have therefore been extensively used to analyze electrification. However, different levels of modeling scope, samples, and methods have led to varying results. There is thus a lack of consensus on under what conditions BETs are competitive with ICETs, and how TCO is affected by different strategic and operational decisions. Subsequently, fleet operators have eschewed TCO models for making strategic decisions regarding freight electrification. A summary of previous literature on the TCO of BETs and electrification is given in Table 1, with some details as follows. High utilization is commonly highlighted as a condition for economic feasibility, allowing BETs to amortize higher fixed costs, such as battery acquisition, with lower variable costs due to the better energy efficiency of electric powertrains (Nykqvist and Olsson, 2021; Feng and Figliozzi, 2013). However, the majority of these studies are based on TCO models for one (representative) vehicle which is then extrapolated to all BETs (Nykqvist and Olsson, 2021; Burke et al., 2022; Macharis et al., 2013; Mareev et al., 2017). This neglects the system level perspective of operating a fleet of vehicles. Furthermore, the majority of these studies only consider aggregate data on networks, shipments, and shipment type. Even if the average mileage and utilization are theoretically viable based on these approaches, this does not necessarily indicate economically viable TCO because the actual utilization is not considered (Taefi et al., 2017).

Some studies develop fleet level models and consider individual utilization of vehicles based on historical data of ICET operations (Link et al., 2021a; Giuliano et al., 2021). By using the operational data of individual ICETs, the TCO of BETs is evaluated by replacing each truck 1-to-1. The benefit of this approach is that the TCO can be estimated more granularly while accounting for the feasibility of electrified operations. However, the implicit assumption that BET fleet routes can be directly mapped onto ICET routes does not fully consider the characteristics and constraints of BETs, and ignores the effects of different routing and scheduling, which could lead to insufficient charging times and inefficient or infeasible driving schedules when accounting for range.

A few studies have therefore also considered optimal routing by incorporating electric vehicle-routing problem (EVRP) optimization in their TCO models (Yang and Hyland, 2024; Schiffer et al., 2021; Al-dal'ain and Celebi, 2021; Giuliano et al., 2021). The methods and constraints in these studies vary, but all have scopes limited to specific case studies, few features, and few scenarios. By

Table 1

A summary of literature on the TCO of BETs.

Paper	Scope	Sample			Method				
	Model level	Transport network	Shipment data	Shipment type	Vehicle utilization	Routing	Sensitivity analysis, Nr of scenarios, and Nr of dimensions		
Macharis et al. (2013)	Vehicle	NC	Agg.	NC	Agg.	NC	Local	<100	<10
Lebeau et al. (2015)	Vehicle	NC	Agg.	NC	Agg.	NC	Local	<100	10
Taefi et al. (2017)	Vehicle	NC	Agg.	NC	Agg.	NC	Local	<100	15
Sripad and Viswanathan (2019)	Vehicle	NC	Agg.	NC	Agg.	NC	Local and Scenario	Unsp.	<10
Tanco et al. (2019)	Vehicle	NC	Agg.	NC	Agg.	NC	Global	Unsp.	<10
Ghandriz et al. (2020)	Vehicle	NC	Agg.	FTL	Agg.	NC	Scenario	<5000	<10
Wolff et al. (2020)	Vehicle	NC	Agg.	NC	Agg.	NC	Scenario	<10	<10
Nykqvist and Olsson (2021)	Vehicle	NC	Agg.	NC	Agg.	NC	Global and Local	<5000	27
Burke et al. (2022)	Vehicle	NC	Agg.	NC	Agg.	NC	Scenario	<100	<10
Tol et al. (2022)	Vehicle	NC	Agg.	NC	Agg.	NC	Scenario	<10	<10
Mauler et al. (2022)	Vehicle	NC	Agg.	FTL	Agg.	NC	Local and Scenario	<100	11
Noll et al. (2022)	Vehicle	NC	Agg.	NC	Agg.	NC	Global	10 000	<10
Mareev et al. (2017)	Vehicle	NC	Agg.	FTL	Agg.	1-to-1 Replacement	Scenario	<10	<10
Liimatainen et al. (2019)	Vehicle	NC	Real	NC	ICET	1-to-1 Replacement	Scenario	<100	<10
Giuliano et al. (2021)	Vehicle	2 Case Studies	Real	FTL	ICET	1-to-1 Replacement	Scenario	<100	<10
Karlsson and Grauers (2023a)	Vehicle	Case study	Real	FTL	ICET	1-to-1 Replacement	Scenario	<10	<10
Link et al. (2021b)	Fleet	Case study	Real	LTL	ICET	1-to-1 Replacement	Scenario	<100	<10
Davis and Figliozzi (2013)	Fleet	Agg.	Agg.	LTL	Agg.	Continuous Approximation	Local and Scenario	<5 000	11
Arroyo et al. (2020)	Fleet	Case study	Unsp.	LTL	Agg.	EVRP	Local	<10	<10
Van Duin et al. (2013)	Fleet	Case study	Real	LTL	BET	EVRP	Scenario	<10	<10
Giuliano et al. (2021)	Fleet	Case study	Real	FTL	BET	EVRP	Scenario	<10	<10
Al-dal'ain and Celebi (2021)	Fleet	2 Networks	Synthetic	LTL	BET	EVRP	Scenario	<10	<10
Schiffer et al. (2021)	Fleet	Case study	Real	LTL	BET	EVRP	Full-Factorial Scenario	<1000	<10
Yang and Hyland (2024)	Fleet	2 Case Studies	Synthetic	LTL	BET	EVRP	Scenario	<100	<10
Our Study	Fleet	110 112 Networks	Real	FTL and LTL	BET	EVRP	Global	110 112	21

Note: NC = not considered, Agg. = aggregated, Unsp. = unspecified.

including simulated utilization and more detailed view of operational decisions and constraints, these studies with EVRP optimization partly address the previous problems, yet are still limited by a distinct lack of generalizability beyond their case studies (Schiffer et al., 2021), particularly in regards to high degrees of uncertainty, sensitivity to complex interactions, and the limited number of scenarios evaluated.

The TCO of BETs is dependent on a wide range of uncertain factors, such as future improvements in battery technology, electricity price volatility, powertrain energy efficiency, and publicly available charging infrastructure. Moreover, these parameters are often aggregated, such as gross battery capacity, state-of-charge (SoC) window, and specific energy consumption (SEC) per kilometer being simplified to one range parameter, despite e.g. battery capacity also affecting the acquisition cost and battery degradation, SoC window affecting the battery lifespan, etc.

Nearly all studies therefore use sensitivity analyses to account for how variation in input parameters might affect TCO in different scenarios. However, none of the studies consider the sensitivity of a wider range of strategic, operational, and exogenous variables,

and particularly not their interactions in conjunction with one another. The focus of these studies are often limited to either strategic or operational decision-making problems. The sensitivity analyses are either scenario based, with non-systematically hand-picked scenarios, or local sensitivity analyses where parameters are varied within a small range and one at a time, thereby not considering the interactions between parameters nor non-linearity. The few global sensitivity analyses in the literature are simple Monte Carlo simulations, which do not systematically analyze the sensitivity of parameters or their interactions, and are therefore less suitable for high degrees of dimensionality (Marchau et al., 2019).

2.3. Deep uncertainty

The complexity of supply chains leads to performance being highly sensitive to initial conditions (Wilding, 1998) and uncertainty (Davis, 1993) which also extends to the subdomain of transportation and route planning (Ukkusuri et al., 2007). The dynamics of uncertainty in supply chains has also been characterized as “deep uncertainty” (Walker et al., 2003).

Previous studies have obtained varying results regarding freight electrification depending on assumptions about uncertain future conditions. Existing sensitivity analyses have not included larger sets of parameters at both at the strategic and operational level to obtain a systematic view. This inhibits the study results to act as a basis for strategic decisions. Decision making under deep uncertainty (DMDU) has seen recent use in transportation planning (Lempert et al., 2022; Jenelius et al., 2023), and could also be applied to commercial electric freight.

Uncertainty in the context of DMDU is distinct from risk as being highly difficult to accurately quantify or estimate. Walker et al. (2003) defines different levels of uncertainty, with deep uncertainty defined as a wide range of outcomes which cannot accurately be modeled stochastically. The DMDU framework includes methods for systematically analyzing such decision problems at a system level (Marchau et al., 2019). Typical approaches investigate complex systems by simulating a wide range of scenarios through computational experiments in order to generate and evaluate a comprehensive set of possible future states based on the models and input parameters (Bankes, 1993; Kwakkel and Pruyt, 2013). The system's relationships and dynamics can thus be studied to identify robust policies (i.e. set of strategic decision variables) given desirable outcomes in the simulated scenarios (Bankes, 1993).

2.4. The vehicle routing problem (VRP)

The vehicle routing problem (VRP) is a well-studied combinatorial optimization problem that involves selecting routes for a fleet of vehicles to optimize time, cost, or other (multi-) objective functions with a set of constraints (Toth and Vigo, 2002). The electric vehicle routing problem (EVRP) is bounded by similar constraints, but with additional considerations related to battery capacity, charging locations, and charging time (Schneider et al., 2014; Montoya et al., 2017; Zhang et al., 2018). Operational decisions of vehicle routing and charging optimization are thus also considered in this study by incorporating EVRP optimization to simulate fleet utilization. This also allows for utilization to be measured more accurately without resorting to aggregated metrics.

2.5. The research gap

Previous literature has covered different aspects of BET feasibility, but not systematically considered the interface between strategic and operational decisions and uncertainty in a generalizable way. Previous TCO studies are therefore limited in their ability to act as a basis for fleet operators to make decisions regarding investments in freight electrification.

This study aims to address this research gap by evaluating BET fleet TCO as a DMDU problem, modeling both strategic and operational decisions, and simulating the utilization of the fleet with EVRP to obtain optimal operational decisions for routing and charging. By systematically exploring over 100 000 scenarios, permutations of strategic decisions and exogenous uncertainties, a comprehensive global sensitivity analysis can be performed to provide insights into conditions for cost-competitive freight electrification.

3. Methodology

3.1. Analytical framework

This study aims to generalize insights into freight electrification beyond case studies by considering deep uncertainty and strategic–operational interfaces, thus providing better understanding of the conditions and strategies where BET fleets are competitive with ICETs. We therefore need to quantify the relative significance of factors affecting the optimal operations and thereby the TCO of BET fleets, particularly relative to their ICET fleet counterparts. An analytical framework is developed to perform a global variance-based sensitivity analysis on a TCO model, based on the DMDU robust decision making model framework XLRM (see Fig. 1), that incorporates strategic decision variables, exogenous uncertainties, and high-resolution simulated utilization of BET and ICET fleets.

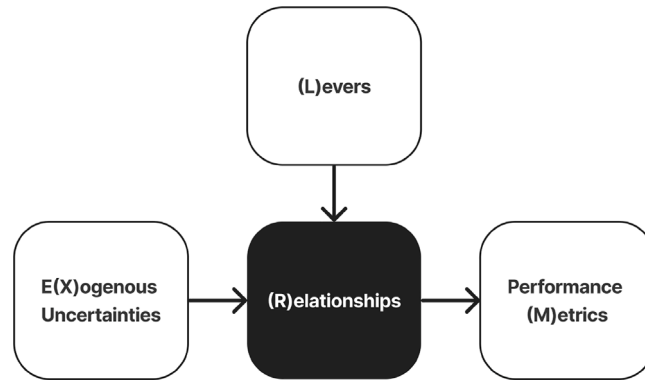


Fig. 1. Conceptual figure of the XLRM Framework, modeling parameters as exogenous uncertainties, decision variables as levers, interactions as relationships, and defining performance metrics for evaluation.

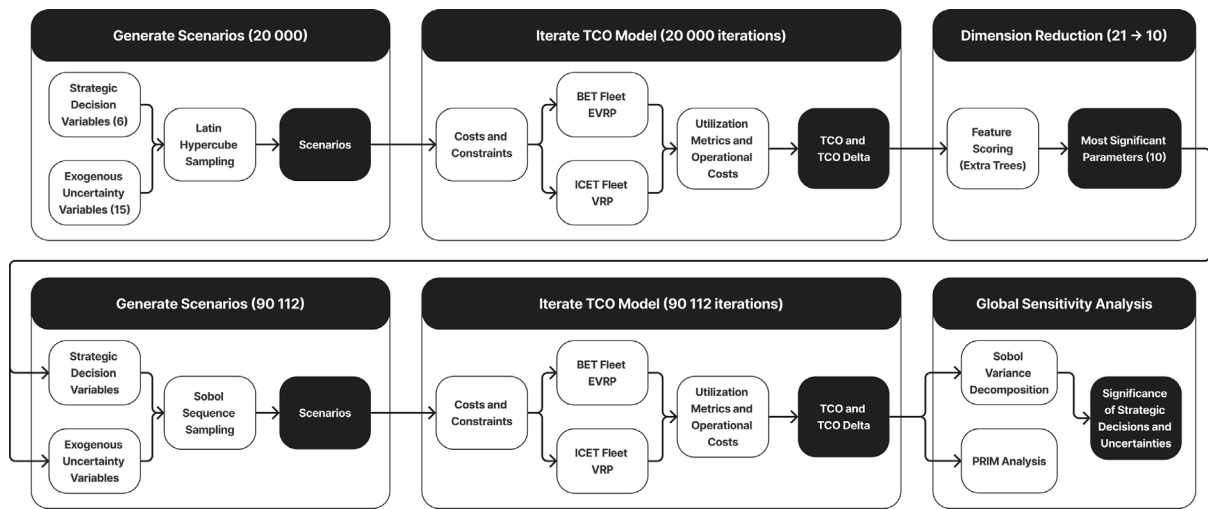


Fig. 2. Research design consisting of input generation, two repetitions of TCO simulation, dimension reduction, and sensitivity analysis.

3.1.1. Research design

To study factors affecting freight electrification, 21 input parameters on TCO, including 6 strategic decision variables and 15 exogenous uncertainty variables, are used to systematically perform a large number of computational experiments.

We generate a large number of scenarios using Quasi-Monte Carlo methods to sample the inputs to the TCO model. The TCO model computes the initial costs and provides constraints for the EVRP optimization which computes the operational performance of the fleets, i.e., high-resolution utilization metrics for the optimal usage of vehicles in the fleet. The EVRP is run twice for each scenario, minimizing the costs of a dynamically sized fleet of BETs and ICETs respectively.

Two sets of scenarios are evaluated. 20 000 scenarios are generated to screen the parameters and reduce the number of dimensions with a feature scoring algorithm Extra Trees. The 10 most significant parameters are then used to generate 90 112 scenarios which are analyzed using Sobol variance decomposition. The procedure is shown in Fig. 2.

By measuring the variance-based sensitivity of each input parameter, the relative significance of strategic decision variables in relation to operations can be quantified. This is realized by mapping the variance of BET fleet TCO, relative TCO between BET and ICET fleets (RTCO), and service proportion of shipments which can be delivered (service level) to variance in each input parameter. Thus, more effective and robust strategic decisions can be made taking into account strategic-operational interfaces and deep uncertainty.

3.1.2. Model framework

The computational TCO model of BET fleets and charging infrastructure (see Fig. 3) is developed according to the DMDU framework XLRM (Lempert et al., 2003), modeling the system with four components: (X) exogenous uncertainties, (L) levers, (R) relationships, and (M) metrics (Marchau et al., 2019). The TCO model includes the operational aspect of optimizing vehicle routes and charging scheduling (EVRP optimization software), and accounts for strategic decision variables and uncertainties.

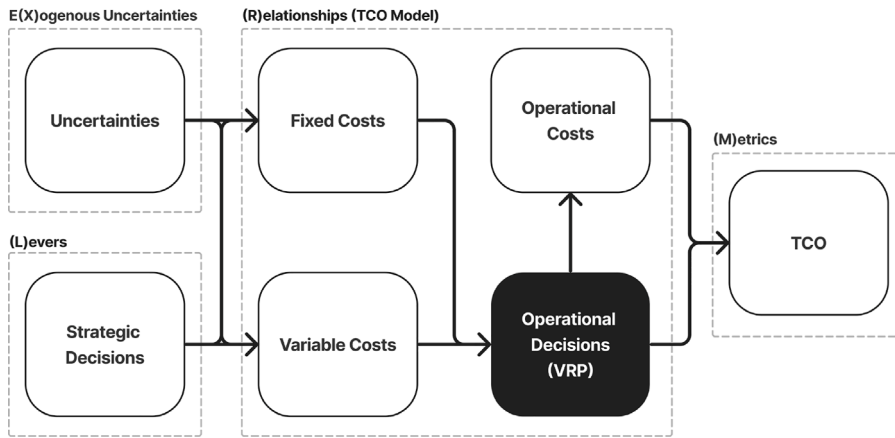


Fig. 3. XLRM model of BET fleet TCO.

Exogenous uncertainties represent the external factors or uncertainties that affect a given system, in our case the variables with a high degree of unpredictability. Particularly salient for electric freight are market dynamics such as wholesale electricity prices and shipment demand, future developments in technology such as battery prices and energy density as well as the topological properties of the transportation network.

Lever represents strategic decisions that a fleet owner-operator can make. The relationships component describes the internal interactions within the system, including feedback loops, dependencies, and interactions between different elements, in this study the dynamic system behavior of the TCO model and EVRP. Lastly, the performance metrics evaluate the effectiveness of the strategic decisions in different scenarios, which in this study are fleet TCO, RTCO, and service level.

3.1.3. Problem setting

The problem is set as follows: a fleet operator makes a set of strategic decisions independently of the uncertainty parameters, then operates their BET or ICET fleet to serve the demand for the network. The fleet operator is an organization that buys, owns, and operates its own fleet of vehicles to serve a given shipment demand. This represents a range of organizations, from contracted freight carriers, to large retail companies that serve their own shipment demand, to third-party logistics providers with their own fleet.

The permutations of exogenous uncertainties describe a given scenario which is then used to benchmark the operational performance of strategic decisions (levers). The variables considered based on literature and expert opinions are given in Table 2. Fleet size is included in the EVRP optimization as a parameter in the cost minimization, and is thus determined after the demand level is known rather than a priori to avoid grossly oversized or undersized fleets. An assumption is therefore that the general level of demand is known beforehand.

Shipment characteristics can vary greatly temporally, spatially, and between customer segments. It is therefore difficult to accurately estimate shipments in terms of weight, volume, distance, and their variance. Thus thousands of real shipment data from various industries were used to generate a data set of shipment demand. The shipment demand for one representative day includes shipment weight, volume (number of pallets), and destination with uniformly distributed delivery time windows, starting between 06:00 to 08:00 and finishing between 12:00 and 20:00. This avoids skewing the computational experiments and ensures accuracy and comparability with real fleet operations.

The charging infrastructure describes the number and location of the chargers in the network. The assumption is that each transport network requires the fleet operator to establish new charging stations and that no other public charging is available. The charging infrastructure decision variables include the maximum charging power of the station and the number of outlets (i.e. how many vehicles can charge simultaneously). When multiple vehicles are charging concurrently, the charging power is split up to the maximum allowed C-rate of each vehicle. Charging infrastructure is always built at the hub, while enroute charging stations, when included, are placed in the network at the halfway point between the hub and the farthest site. Additionally, charging at the loading bay is possible if a charging outlet is available.

3.1.4. Sampling method

Due to the high number of dimensions in the input to the model, random uniform sampling of each individual parameter does not lead to uniformity among each permutation of input parameters, and is therefore insufficient to systematically measure the sensitivity of a system of this complexity (Marchau et al., 2019). To ensure a parameter space with enough coverage, the input parameters must be sampled to obtain additional uniformity properties. Thus, Latin Hypercube and Sobol sampling are used to generate uniformly distributed sequences of each exogenous uncertainty and strategic decision variable (lever) which permutated together form scenarios for each computational experiment. The sampling is performed in the two scenario generation steps shown in Fig. 2 with Latin Hypercube sampling and Sobol sampling respectively, which are often used in DMDU analyses (Marchau et al., 2019; Helton and Davis, 2003).

Table 2

Parameters affecting freight electrification divided into strategic costs, operational costs, and operational constraints. Parameters which are characterized by deep uncertainty according to industry experts and parameters that are decision variables for fleet owner-operators are marked as such.

Category	Parameter	Deep uncertainty	Decision variable
Strategic costs	Vehicle acquisition cost		
	Vehicle residual value		
	Fleet utilization	x	
	Capital costs	x	
	Battery cost (per kWh)	x	
	Battery capacity (kWh)		x
	Battery degradation	x	
	Battery residual value	x	
	Charging infrastructure cost	x	
	Nr of charging stations		x
	Nr of charger outlets		x
	Charger power (kW)		x
	Charging infrastructure holding period		
	Charging infrastructure maintenance cost		
Operational costs	Electricity price	x	
	Driver wages		
	Energy efficiency (Specific Energy Consumption)	x	
	Vehicle maintenance costs	x	
	Tire costs		
Operational constraints	Battery capacity (kWh)		x
	Battery energy density (kg/kWh)	x	
	Vehicle weight		
	Vehicle weight class		
	EV gross vehicle weight limit	x	
	State-of-Charge (SOC) Window (max/min %)		x
	Maximum charging-rate (C)		x
	Network size	x	
	Shipment demand	x	
	Shipping mode (FTL or LTL)	x	

3.1.5. EVRP optimization

This study uses EVRP optimization software developed by [Einride \(2024\)](#), a Swedish transportation technology company specialized in electric and autonomous freight, for the commercial operation and planning of freight electrification. The proprietary software optimizes fleet size, shipment allocation, routing and charging utilization to meet the demand of a set of shipments for a given transport network. The charge- and discharge-cycles obtained from the simulation allow battery aging to be calculated at a more granular level so that the depreciation cost of the battery can be correctly represented in the TCO model by using nominal values of battery depreciation as an input. Once the optimal utilization and fleet size is computed, the TCO is recalculated and updated to reflect the hastened or slowed depreciation of the battery for each vehicle.

3.1.6. Analytical methods

The analysis aims to investigate the significance of strategic decisions and uncertainties while considering strategic–operational interfaces by evaluating the statistical relations between the input parameters and output metrics for BET fleet TCO and RTCO.

Thus a global sensitivity analysis is performed in two steps as shown in [Fig. 2](#). Firstly, due to computational complexity, feature scoring reduces the number of free dimensions by approximating the relative significance of the input parameters and is performed using the machine learning algorithm extra random trees on 20 000 scenarios to approximate the total order results of the following Sobol analysis ([Jaxa-Rozen and Kwakkel, 2018](#); [Geurts et al., 2006](#)). Secondly, Sobol variance decomposition measures the significance of the input parameters and the significance of their interaction effects ([Pianosi et al., 2016](#)).

This study uses the implementation of Sobol in the EMA workbench (based on SALib) which requires $N \times (2 \times D + 2)$ samples to compute first and second order indices, where D is the number of free dimensions and N is a power of 2. With $D = 10$ and $N = 4096$, 90 112 sets of input parameters are generated, each corresponding to a scenario. Additional analyses are performed using the Patient Rule Induction Method (PRIM) to identify parameter regions and sub-regions where a target variable assumes a relatively high or low value ([Friedman and Fisher, 1999](#)). A region found through PRIM has two properties, density and coverage. Density describes how many of the scenarios within the region assume a certain value, i.e. the probability of a randomly selected scenario within that region fulfilling that condition. Coverage describes how many of the total scenarios assuming a certain value are found within that region, i.e. the probability of a randomly selected scenario that fulfills the condition being within that region.

Table 3

List of exogenous uncertainty (X) parameters for BETS. All parameters are scalar unless specified as categorical (c) or discrete (i). Values according to industry experts are marked “IE”.

Type	Notation	Parameter	Min	Max	Unit	Source
X	$b_{density}$	Battery energy density	0.15	0.3	$\frac{kWh}{kg}$	IE
X	$c_{hardware}$	Charging station hardware cost	100	150	$\frac{kSEK}{n_{outlets}}$	IE
X	c_{kWh}	Battery cost per kWh	1000	3000	$\frac{SEK}{kWh}$	IE, Nykvist and Olsson (2021)
X	c_{power}	Charging station power capacity cost	2500	4500	$\frac{SEK}{kW}$	IE
X	SEC_{curb}	Specific energy consumption at curb weight (energy efficiency)	0.8	1.5	$\frac{kWh}{km}$	IE
Xc	M_{EU}	EV gross vehicle weight limit modifier	0	2	tons	Council of European Union (2019)
X	$c_{maintenance}$	BET maintenance costs	40	100	%	Rogstadius (2022)
Xi	$n_{shipments}$	Number of shipments	1	2	$\frac{n}{site}$	
Xi	n_{sites}	Number of sites	1	20		
X	p_{kWh}	Wholesale electricity price	0.5	3	$\frac{SEK}{kWh}$	IE
Xc	$r_{battery}$	Battery residual value at EOL	10	55	%	Rogstadius (2022) and Helfer (2023)
X	$R_{network}$	Network radius	20	250	km	
Xc	$type_{network}$	Shipping mode	FTL	LTL		
X	ρ	Internal interest rate	6	12	%	IE
Xi	$\Gamma_{vehicle}$	Vehicle utilization (yearly)	200	250	days	IE

Table 4

List of lever (L) parameters for BETs. All parameters are scalar unless specified as categorical (c) or discrete (i). Values according to industry experts are marked “IE”.

Type	Notation	Parameter	Min	Max	Unit	Source
L	$b_{capacity}$	Gross battery capacity	200	1000	kWh	IE
L	C_{max}	maximum C-rate	0.5	2	$\frac{1}{h}$	IE
Lc	CI	Charging infrastructure strategy	Hub	Enroute		
Li	$n_{outlets}$	Number of Outlets per Charging Station	1	8		IE
L	SOC_{min}	SoC minimum	10	50	%	IE
L	W_{max}	Maximum charger power	100	1000	kW	IE

3.2. Model formulation

3.2.1. Input data

The parameter settings are based on values in literature and expert opinions from Einride (2024). In accordance with the DMDU framework, no assumptions are made regarding the probability distributions of the parameters in order to study the edge cases to hedge against deep uncertainty and support robust decision making. The ranges used in the study are given for exogenous uncertainties in Table 3, strategic decision variables (levers) in Table 4, and other model constants in Table 5. Cost parameters are given in SEK (approximately 1 SEK = 0.1 USD).

3.2.2. TCO model

The TCO model considers strategic, operational, and capital costs described in Eq. (1) where TCO is divided by the number of days the fleet is active throughout its lifespan. SC is the fixed cost for each vehicle i and charging station j , CC_i is the capital cost, OC_i is the daily operational cost for each vehicle i as computed by the EVRP and charging station j according to the yearly maintenance cost.

$$TCO = \sum_i (SC_i + CC_i + OC_i) + \sum_j (SC_j + OC_j), \forall i \in \{vehicles\}, j \in \{charging\ stations\} \quad (1)$$

The fixed costs for each vehicle per day is given by Eq. (2) with the acquisition cost AC for the vehicle excluding battery (e.g. chassis and powertrain). The acquisition cost of the battery is the battery capacity $b_{capacity}$ multiplied with the battery cost per kWh c_{kWh} . The fleet utilization Γ_{fleet} is the number of days per year that the fleet is utilized, and $Y_{vehicle}$ is the holding period of the vehicle excluding battery. $Y_{battery(nominal)}$ is the nominal lifespan of the battery in years which is updated after the EVRP has simulated the predicted lifespan based on charge and discharge subcycles. r is the respective residual value of the vehicle and battery as a percentage of the acquisition cost.

$$SC_i = \frac{AC_{vehicle}(1 - r_{vehicle})}{Y_{vehicle} \Gamma_{fleet}} + \frac{c_{kWh} b_{capacity}(1 - r_{battery})}{Y_{battery(nominal)} \Gamma_{fleet}}, i \in \{vehicles\} \quad (2)$$

The mean yearly capital cost of each vehicle in Eq. (3) is calculated as rate ρ with a linear depreciation in the value of the asset at an estimated half of the acquisition cost according to the (Swedish Transport Administration, 2023).

$$CC_i = \frac{1}{2} \rho AC_{vehicle}, i \in \{vehicles\} \quad (3)$$

Table 5

List of constant (C) parameters. Values according to industry experts are marked “IE”.

Type	Notation	Parameter	Value	Unit	Source
C	$c_{rejection}$	Rejection cost per shipment	4000	SEK	IE
C	$c_{charging\ maint.}$	Charging station maintenance cost	50 000	$\frac{\text{SEK}}{\text{year}}$	IE
C	c_{driver}	Driver wages	340	$\frac{\text{SEK}}{\text{h}}$	Rogstadius (2022)
C	$cycles_{nominal}$	Nominal life battery (cycles)	1500	FCE	Rogstadius (2022)
C	SEC_{cargo}	Specific energy consumption per kg payload	145	$\frac{\text{J}}{\text{kgokm}}$	IE
C	α_j	Enroute capacity cost modifier	200	%	IE
C	$c_{maintenance}$ (ICET)	ICET maintenance costs	0.12	$\frac{\text{SEK}}{\text{km}}$	Rogstadius (2022)
C	$Y_{battery (nominal)}$	Nominal life battery	10	years	IE
C	SOC_{max}	Battery SoC maximum	90	%	IE
C	c_{tires}	Tire costs	0.006	$\frac{\text{SEK}}{\text{km}}$	Rogstadius (2022)
C	$r_{vehicle}$	Vehicle residual ratio	9	%	Rogstadius (2022)
C	WC	Vehicle weight class	40		
C	$AC_{vehicle}$	Acquisition cost chassis and powertrain	2 048 660	SEK	Rogstadius (2022)
C	$Y_{chargers}$	Holding period charging stations	5	years	IE
C	a_1	DoD degradation logarithmic parameter	-38975.32		Adapted from (Helfer, 2023)
C	a_2	DoD degradation logarithmic parameter	-5299.02		
C	b_1	DoD degradation constant parameter	-34292.6		
C	b_2	DoD degradation constant parameter	392.18		

The variable cost of each vehicle, as computed by the EVRP optimization, consists of the total specific energy consumption SEC , maintenance costs, and tire costs as a function of the total distance driven d_i , and the driver wages as a function of total time spent driving, loading, unloading, and charging t_i . Total specific energy consumption represents the energy consumed per distance driven, and a larger or lower SEC can also be used to proxy for the effect of temperature on energy consumption.

$$OC_i = (p_{kWh} \sum SEC + c_{maintenance} + c_{tires}) d_i + c_{driver} t_i, i \in \{vehicles\} \quad (4)$$

The fixed cost of building a charging station is given by Eq. (5) and is a function of the cost per kWh $power$ and maximum power that the station provides W_{max} , as well as the cost $c_{hardware}$ of each outlet $n_{outlets}$. The cost is then divided by the holding period of the charging station $Y_{chargers}$ and Γ_{fleet} . The costs represent acquisition, depreciation, grid connection, grid subscriptions, electric installation, and site construction. If an enroute charging station exists, the cost is increased by a factor $\alpha_j = \{2 \text{ if } enroute, \text{ else } 1\}$ to represent higher costs and additional land leasing costs.

$$SC_j + CC_j = \alpha_j \frac{c_{power} \times W_{max} + c_{hardware} \times n_{outlets}}{Y_{chargers} \Gamma_{fleet}}, j \in \{charging\ stations\} \quad (5)$$

The variable cost for maintaining the charging infrastructure is based on a yearly maintenance cost divided by Γ_{fleet} , with the assumption that it is only used privately.

$$OC_j = \frac{c_{charging\ maintenance}}{\Gamma_{fleet}}, j \in \{charging\ stations\} \quad (6)$$

3.2.3. Battery depreciation

A large portion of the acquisition costs for BETs is due to the battery, and is also the primary reason why BETs are more expensive than ICETs. However, the lifecycle costs are difficult to accurately estimate due to complex chemistries and non-linear behavior. The lifecycle cost of the battery depends on the vehicle's utilization, and is therefore unknown a priori, despite being an input to the EVRP optimization. To solve this, nominal values for battery lifecycles are first used to compute the costs used by the EVRP optimization; measured in number of cycles and number of years.

The depreciation of the battery is thereafter updated according to the computed lifespan based on the simulated utilization from the EVRP output in Eq. (7), considering the predicted number of cycles $cycles_{predicted}$ in the battery's life (based on utilization), the simulated charge and discharge cycles for daily operations obtained from the EVRP $cycles_{vrp}$ and the yearly utilization of the fleet Γ_{fleet} .

$$Y_{battery} = \frac{cycles_{predicted}}{cycles_{vrp} \times \Gamma_{fleet}} \quad (7)$$

There are a variety of mechanisms affecting battery aging, or more properly “state-of-health (SoH) degradation”, during charging, discharging, and idling. These are further influenced by factors such as temperature, depth-of-discharge (DoD), current, cut-off voltage, and state-of-charge (SoC) (Guo et al., 2021). The main factors affecting battery cost in the context of fleet transportation, how fast the battery is charged and how much of the battery is depleted between charges, are included in this study, while other factors such as different battery chemistries are excluded.

Table 6

Selected parameters to be sampled after feature scoring.

Uncertainty parameters (X)	Lever parameters (L)
Network radius	Gross battery capacity (kWh)
Number of sites	SoC Minimum (%)
Shipping Mode (FTL or LTL)	Nr of charging stations
Specific energy consumption (kWh/km)	Charger power (kW)
Fleet utilization (days per year)	
Wholesale electricity price (€/kWh)	

Table 7

Constant values for the parameters that were excluded from the second stage before the Sobol analysis.

Fixed values for non-sampled parameters (Sobol)			
Parameter	Former type	Fixed value	Unit
Number of shipments per site	X	2	
Number of outlets per charging station	L	4	
Charging hardware cost	X	3500	$\frac{\text{SEK}}{\text{kW}}$
Charging outlet cost	X	130 000	$\frac{\text{SEK}}{n_{\text{outlets}}}$
Maintenance cost ratio BET and ICET	X	80%	
Interest rate	X	5%	
Battery cost per kWh	X	2000	SEK
Battery residual value	X	10%	
Maximum C-rate	L	2	C
EU gross vehicle weight limit	X	2	tonnes
Battery energy density	X	0.18	$\frac{\text{kWh}}{\text{kg}}$

The lifespan of the battery as measured in cycles is normalized to full cycle equivalents (FCE), i.e. a complete 100% charge and discharge of the battery's capacity. The predicted lifespan of the battery is based on the DoD, i.e. how much of the battery's SoC is depleted between charges (see Eq. (8) adapted from [Helfer \(2023\)](#)).

$$cycles_{predicted}(DoD) = \begin{cases} a_1 \times \log(DoD) + b_1 & \text{if } 10\% \leq DoD < 35\% \\ a_2 \times \log(DoD) + b_2 & \text{if } 35\% \leq DoD \leq 90\% \end{cases} \quad (8)$$

To convert the cycle life to a yearly lifespan, the charge cycles simulated for each vehicle is normalized and summed in Eq. (9) to obtain the yearly expenditure of cycles during operations.

$$cycles_{vvp} = \sum_k a_{cycle, k}(J_{charged}, C_{rate}), \forall k \in \{charge\ subcycle\} \quad (9)$$

Charging at high charging-rates (C-rates), more commonly known as fast charging, generates more heat, increased internal resistance, and agitates the internal chemical processes in the battery, reducing its battery capacity and effective cycle-life ([Guo et al., 2021](#); [Rogstadius, 2022](#)). To account for this, Eq. (10) adapted from [Rogstadius \(2022\)](#) is used with a nominal C-rate of 1.

$$a_{cycle}(J_{charged}, C_{rate}) = 0.5 \times J_{charged} \times \max(0.75, (\frac{C_{rate}}{C_{nominal}})^2) \quad (10)$$

The C-rate $C_{i,j}$ for each vehicle's charge cycle is computed by measuring each simulated charge-cycle k and comparing the effective charger power used $W_{i,k}$ to the battery capacity $b_{capacity}$ of the vehicle i (Eq. (11)).

$$C_{i,j} = \frac{W_{i,k}}{b_{capacity}}, i \in \{vehicles\}, k \in \{charge\ subcycle\} \quad (11)$$

4. Results

As mentioned before, feature scoring is first used to reduce the number of free dimensions from 21 to 10. Additionally, a large number of iterations are used to measure the sensitivity of each parameter, ensuring that the input space is covered, and to compute both first and second order Sobol indices. All computations were performed on an Apple M2 8-core processor with computation times of 37 h for the 40 000 EVRP instances for feature scoring and 165 h for the 180 224 EVRP instances for the final sensitivity analysis.

More specifically, parameters are screened using feature scoring of service level (1 - rejection rate), TCO, TCO normalized per shipment, relative TCO for BETs and ICETs (RTCO), and RTCO normalized per shipment (See Figure A.10 in Appendix A). The 10 most significant of the 21 sampled parameters are selected for the second repetition of simulations, and are given in [Table 6](#). The remaining parameters are assigned constant values (as given in [Table 7](#)). In the following sections, the total effect index describes the relative significance of each input parameter on a given output metric. The first and second order indices describe the proportion of output variance that can be explained by variance in an individual or interaction between a pair of input parameters respectively.

Table 8

Five most significant parameters in order of effect on TCO per shipment.

Rank	Parameter	Total order effect	Confidence band
1	Number of sites	0.898510	0.112013
2	Network radius	0.626226	0.109778
3	Gross battery capacity	0.583632	0.084009
4	nr of charging stations	0.418862	0.073309
5	Specific energy consumption	0.399839	0.075746

Table 9

Sobol first and second order indices in descending order of significance. Second order interaction effects are indicated by two parameters in parentheses.

Rank	Parameter	Effect	Confidence band
1	Number of sites	0.166740	0.056694
2	(gross battery capacity, number of sites)	0.106503	0.085526
3	(network radius, number of sites)	0.087997	0.072285
4	(nr of charging stations, number of sites)	0.086962	0.073788
5	(number of sites, fleet utilization)	0.084834	0.084607

Table 10

Five most significant parameters in order of effect on RTCO per shipment.

Rank	Parameter	Effect	Confidence band
1	Number of sites	0.975229	0.138308
2	Gross battery capacity	0.678885	0.113197
3	Network radius	0.614530	0.099076
4	nr of charging stations	0.460019	0.085670
5	Specific energy consumption	0.454881	0.097249

4.1. BET fleet TCO

As shown in Table 8, TCO, normalized per shipment, is predominantly affected by the number of sites; i.e. the size of the network (number of sites and shipments), with significant impact from network radius (maximum distance between hub and sites) and battery capacity. The most significant isolated parameter is shipment demand (see Table 9), followed by its interaction effects (see Table 9), thus indicating that network features are more significant than vehicle features, which has seen lacking attention in the literature (see Table 1). The remaining variance has too low confidence to be allocated to any other first or second order effect, indicating that the TCO of a fleet of BETs is sensitive to noise caused by e.g. network topology or that higher order interactions dominate.

4.2. RTCO

By decomposing the variance of the RTCO between BET and ICET fleets for the same scenario, we identify the conditions for cost-competitive freight electrification. Table 10 shows that the order and magnitude is similar to that of BET fleet TCO; with number of sites, battery capacity and network radius clearly most significant.

The results of the sensitivity analysis are similar due to correlation between variance in BET fleet TCO and variance in the resulting RTCO between BET and ICET fleets. Interestingly, the interaction effects between the top three parameters are ranked highly among the first and second order Sobol indices and, although the confidence bands are too wide to be conclusive, still indicate that the interdependencies between these three are significant when determining cost-competitive electrification (see Table 11).

Furthermore, sensitivity to electricity price is low, indicating its volatility and unpredictability is not necessarily a barrier to BET adoption. This is in contrast to Nykvist and Olsson (2021), Mauler et al. (2022), and Noll et al. (2022) which identify energy prices as important. These studies however consider aggregated utilization metrics, while the most significant factors identified in this study are related to fleet operations rather than market prices. The interactions between network properties and operational range clearly dominate all other factors, indicating that lower total fuel costs are more sensitive to operational efficiency than energy prices. This is corroborated by Yang and Hyland (2024) who also use EVRP optimization and note that electrification rates, while influenced by the relative cost per mile of BETs and ICETs (including electricity price), are more influenced by how much demand can be served by BETs.

4.3. Service level

The Sobol variance decomposition of service level shows that service level is primarily affected by two parameters: battery capacity and network radius. Network radius and battery capacity are clearly most significant, as they affect the range requirements for serving the demand and the range of the vehicles respectively. However, a large portion of the variance is still not caused by variance in network radius, battery capacity, nor any of the other parameters. This indicates that the remaining variance in service level might be caused by other factors such as the spatio-temporal distribution of the demand and sites in the network (see Tables 12 and 13).

Table 11

Sobol first and second order (indicated in parentheses) indices.

Rank	Parameter	Effect	Confidence band
1	Number of sites	0.159334	0.054439
2	(gross battery capacity, number of sites)	0.139868	0.097879
3	(network radius, number of sites)	0.090784	0.071766
4	(gross battery capacity, network radius)	0.084919	0.092334
5	(gross battery capacity, SoC minimum)	0.076890	0.079665

Table 12

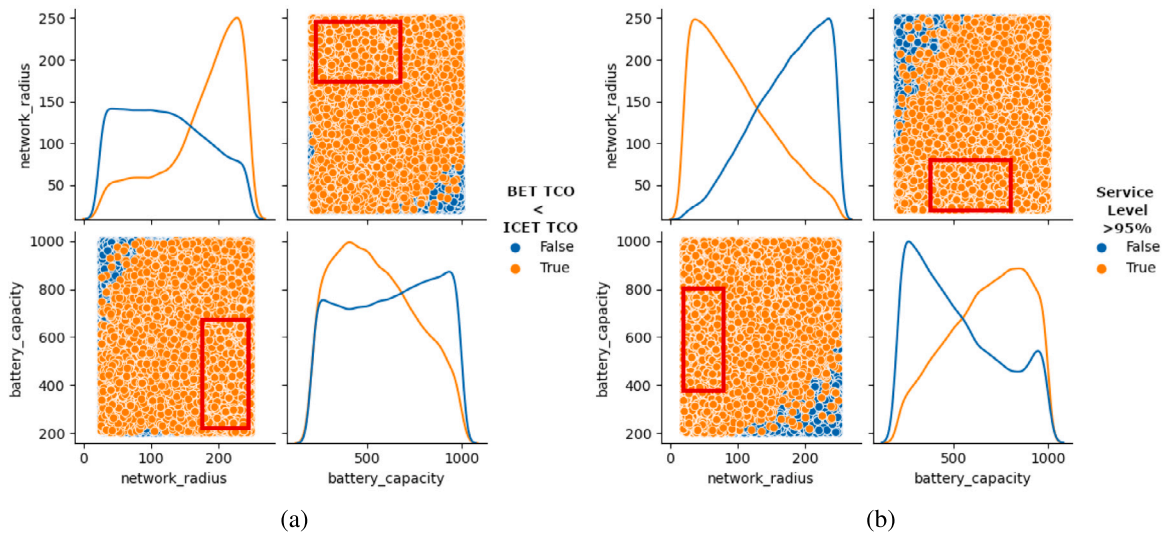
Five most significant parameters in order of total effect on service level.

Rank	Parameter	Total Order Effect	Confidence band
1	Network radius	0.625364	0.032019
2	Gross battery capacity	0.557693	0.03416
3	SoC minimum	0.214413	0.018989
4	Specific energy consumption	0.175435	0.01412
5	Number of sites	0.151658	0.013249

Table 13

Sobol first and second order indices for service level (rejection rate) in descending order. Second order interaction effects are indicated by two parameters in parentheses.

Rank	Parameter	Effect	Confidence band
1	Network radius	0.306733	0.033188
2	Gross battery capacity	0.276969	0.035732
3	(gross battery capacity, network radius)	0.098151	0.047821
4	SoC minimum	0.046058	0.021521
5	(network radius, SoC minimum)	0.042186	0.042255

**Fig. 4.** Regions identified with PRIM analysis bounded by network radius and battery capacity where (a) BET fleets have a lower TCO than ICETs and (b) have a service level above 95%.

4.4. PRIM analysis

Scenarios where BET fleets have a lower TCO than ICET fleets are bounded by the distances between sites and shipment demand, with distances above 120 km covering 80% of such cases. The scenarios where BET fleets are most likely to be cost-competitive (see Fig. 4a) have distances between 175 and 245 km, battery capacities between 220 and 675 kWh, and at least 14 shipments at 7 sites, leading to 75% of the scenarios within this region being cost-competitive, although only including 16% of all cost-competitive scenarios. This includes both fully (100% service level) and partially electrified scenarios.

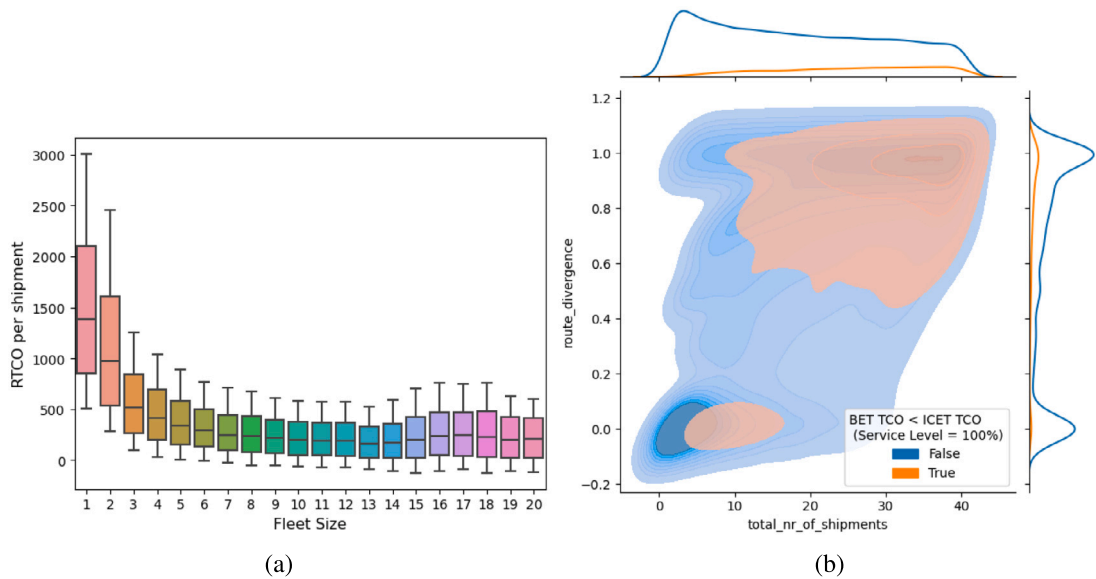


Fig. 5. (a) Mean RTCO (in SEK) for fleets with a service level above 95% as fleet size increases as a result of higher shipment demand. (b) Bivariate kernel density estimation of shipment demand and route divergence for scenarios with 100% service level. Peak probability for cost-competitive TCO at high demand and high route divergence.

Conversely, PRIM analysis of service levels shows that battery capacities above 350 kWh and distances below 160 km are needed to ensure service levels above 95%, with both high coverage (representativeness) and density (probability) at 75% and 80% respectively (see Fig. 4b). This trade-off between cost-competitiveness and high service levels indicates that full electrification is not robust under deep uncertainty. Many networks can become more cost-efficient through electrification of a subset of the shipment demand served by a network.

5. Discussion

5.1. BET competitiveness scales with demand and fleet size

Total shipment demand is shown to be most significant for both the TCO of BET fleets as well as their RTCO with ICETs. Cases where batteries were grossly oversized led to poor cost-performance per vehicle being inflated with fleet size. Interestingly however, higher demand positively correlates with the probability of $RTCO < 0$, i.e. cost-competitiveness, when batteries are appropriately sized (see Fig. 5a).

More shipments and greater distances lead to more kilometers driven per vehicle and higher utilization, thus amortizing the fixed costs, in line with previous studies (Noll et al., 2022). More significantly however is that as the fleets grow larger, the BETs achieve an emergent economies of scale effect in the operational planning that mitigates their operational constraints.

As fleet size scales with demand, EVRP optimization enabled balancing shipment and energy demand across the fleet to more evenly utilize the vehicles. This pooling effect minimizes charging and wait times and allocates shipments more efficiently, thereby lowering wage costs (wait times) and reducing the number of rejected shipments bounded by time windows and their associated rejection costs.

With higher demand and more vehicles, the EVRP is more likely to find operational optimizations that change the routes and charging schedules, leading to increasingly different planning between BETs and ICETs as noted in Section 5.3, with clear correlations between higher shipment demand, route divergence, and lower RTCO. This can be seen in Fig. 5(b) with a clear peak in the probability of BETs having a lower TCO than ICETs at higher demand and route divergence. Increasingly different operations between BET and ICET fleets can be seen by the absence of low route divergence metrics at high demand. This has significant implications for fleet operators, showing that as electrification increases the characteristics of the operational decisions will change more dramatically (compared with ICETs) and require more appropriate planning for BETs to be competitive. Furthermore, strategic decisions need to take this operational aspect into account.

5.2. Trade-off between TCO and service level: Robust electrification strategies

In scenarios where the BET fleet was unable to deliver all shipments, the penalty cost per rejected shipment can represent the cost of the shipment being delivered by an ICET instead. Since strategic decisions are sampled independently of the uncertainty

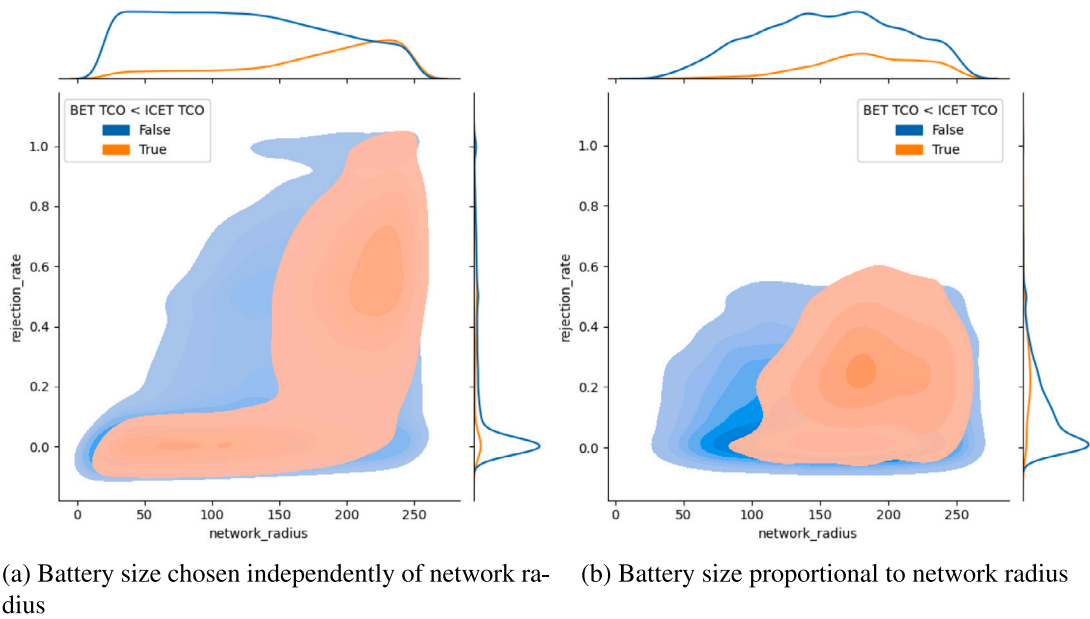


Fig. 6. Bivariate kernel density estimation of scenarios where the BET fleet had a lower TCO than the ICET fleet, where (a) includes all scenarios while (b) is filtered to only include scenarios where the BETs have sufficient range. Darker shades indicate a higher probability density.

parameters, a large proportion of the scenarios consider fleets of BETs where the vehicles have undersized batteries and therefore have insufficient range. These scenarios then show the cost-competitiveness of partial electrification.

When considering all scenarios, partial electrification in scenarios with service levels as low as 20% is particularly cost-competitive at larger distances above 150 km, with an optimal service level (i.e. electrification level) of 25%–50% (see Fig. 6a). In these scenarios the shorter range shipments are served by a fleet with smaller and more affordable batteries while the farther shipments are served by ICETs. Full electrification on the other hand is most likely to be cost-competitive between 50 and 120 km.

The strategic decisions concerning fleet procurement can be made with or without prior knowledge of the network structure and ranges. Fig. 6a suggests that, if the range is unknown, that 25 to 50% of the fleet could be electrified while ensuring a higher likelihood of competitiveness. If the network's range requirements are known in advance, the battery size can be chosen accordingly. Fig. 6b shows the subset of scenarios where a battery capacity was chosen with sufficient range to serve the entire network. Even with sufficient range, partial electrification at 70 to 80% is still most likely to be competitive. Interestingly however, longer ranges at over 150 km become more competitive, suggesting that predictable network structures allow for electrification at longer distances.

This indicates that electrifying up to 100% of short haul (≤ 300 km round-trip) and up to 75% of regional haul trucking (≤ 500 km round-trip) can be reliably cost-competitive over a wide range of scenarios given that robust strategic and operational decisions are made taking into account operational constraints such as routing, charging scheduling and prior knowledge about network structure.

5.3. Significance of operational decisions and EVRP optimization

The effect of planning on electrification is an important consideration for fleet operators and a major gap in the literature (Gillström et al., 2024). Therefore, to measure the significance of EVRP optimization on TCO, BET and ICET fleets are compared by measuring how much their planning of routes, charging, and scheduling differ. The difference in planning is measured with route divergence; the ratio of BETs that drive the same set of routes as an ICET counterpart (excluding detours for charging). The impact of EVRP optimization for BETs is shown to be highly significant for cost-optimization on the fleet level, with significant divergence (see Fig. 7a) with 87.03% of the BET fleets diverging to some extent (see Fig. 7b).

When swapping a planned shipment from one truck to another in order to create a more feasible or cost-efficient route plan for BETs, the other truck will also more likely need to reorganize and reallocate its shipments with other trucks in the fleet, and so on. This can be obviously seen in Fig. 7a where route divergence is clearly skewed towards 100% and 0%.

The mean route divergence of 52.35% shows that over half of all BETs across all scenarios must drive a different set of routes than ICETs in order to be cost-optimal. In other words, nearly 90% of all BET fleets would have sub-optimal utilization and TCO if ICET routes were used (see Fig. 7b). In conclusion, EVRP optimization is highly significant for reducing the TCO of BETs, and the operational decisions need to be taken into account when making strategic decisions.

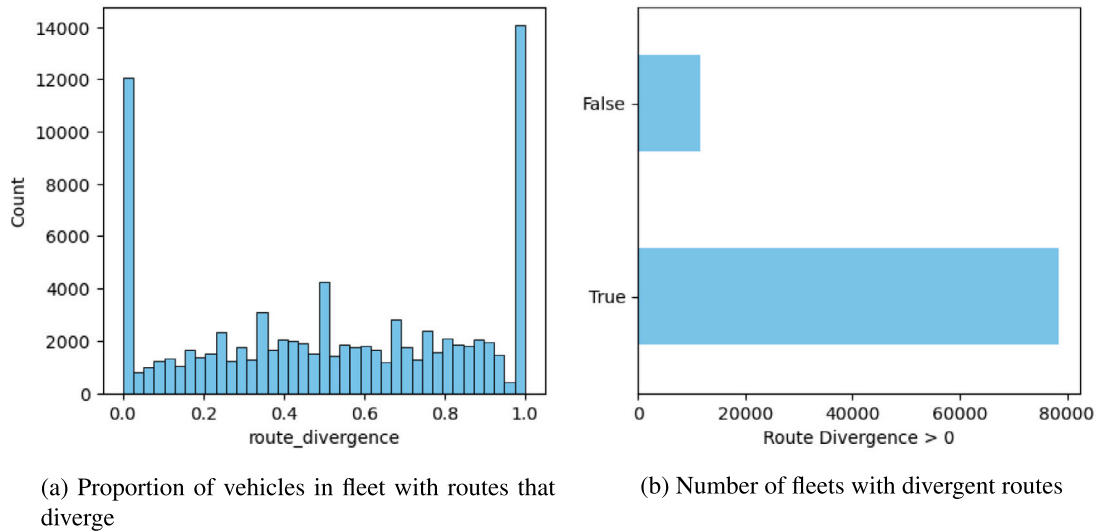


Fig. 7. Number of scenarios where the BET fleet has a diverging set of optimal routes compared with the ICET fleet, measured as the proportion of vehicles in the fleet that have a different set of optimal paths in (a) and the number of fleets with any divergence in (b).

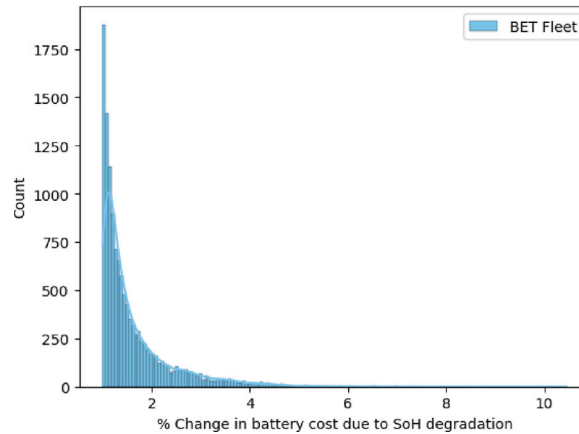


Fig. 8. Histogram of percentual change in battery depreciation cost due to battery SoH degradation. Cases where no change in cost occurs are filtered.

5.4. Battery degradation

The effect of battery SoH degradation on depreciation costs was not significant even when considering fast enroute charging, only occurring in 14.14% of the results. The change in cost is also marginal, falling within 0 and 4% with edge cases going up to 10% (see Fig. 8). Considering that the mean battery cost constitutes 9.28% of the TCO, the worst case increase of 10% still amounts to less than a 1% change in TCO.

Conclusively, fast charging only causes a marginal increase in TCO, which is outweighed by the benefit of shorter charging durations. This is partly due to the size of the vehicles and their batteries, because moderate C-rates can be achieved even with high powered chargers if the battery capacity is high enough which is typically the case for 40-tonne tractor-trailer combinations.

Accelerated battery degradation only became noticeable when the battery consistently has high cycle depths (see Fig. 9a). The highest battery degradation costs however arise from a combination of both high cycle depths and high charger power (see Fig. 9b). While not as significant as other factors such as demand and distance, battery SoH degradation can be a relevant factor for networks where the vehicles consistently deplete most of their battery capacity between charges and charge with high C-rates, but can otherwise be omitted from TCO models.

6. Conclusion

This paper proposes and applies a novel analytical framework for analyzing heavy-duty freight electrification that takes into account strategic decisions, operational decisions, deep uncertainty, and strategic-operational interfaces, which has been lacking

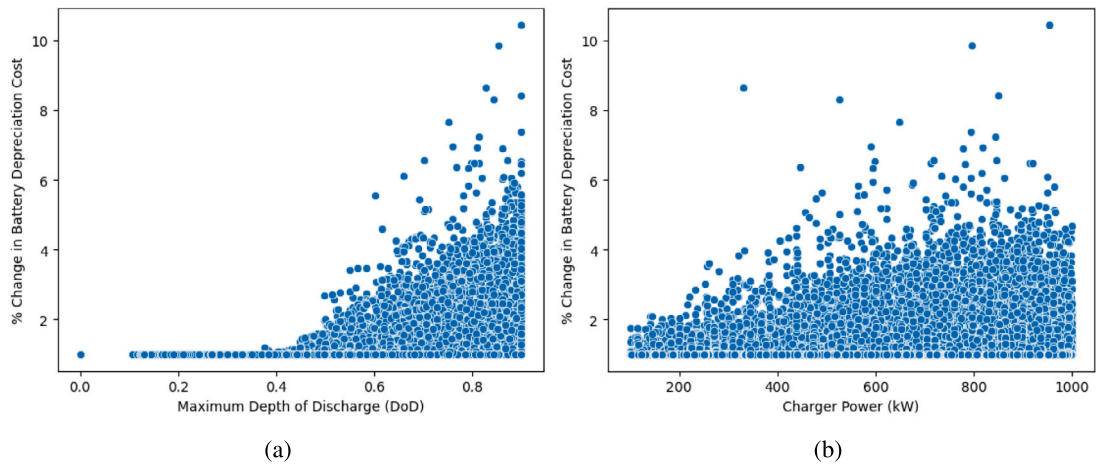


Fig. 9. (a) Percentual change in battery depreciation cost plotted against maximum depth of discharge, i.e. cycle depth. Battery costs start to increase once the battery cycles approach a full FCE. (b) Percentual change in battery depreciation cost plotted against charger power.

in the literature. By systematically considering 110 112 scenarios, we identify factors that significantly affect freight electrification. The results can be generalized across a wide range of freight electrification scenarios and contribute to the decarbonization of road freight. The new findings include the following.

Operational efficiency is the key enabler of freight electrification. While previous studies on the technical and economical feasibility of BETs varyingly indicate that battery capacity (Yang and Hyland, 2024), charging (Giuliano et al., 2021), battery prices (Nykvist and Olsson, 2021), and/or energy prices (Mauler et al., 2022; Noll et al., 2022) are the most significant, our study finds that the competitiveness of electrification is instead strongly dominated by factors that influence operational efficiency, which becomes a prerequisite for electrification before needing to consider other factors. Larger networks served by larger fleets have an economies of scale effect, since larger fleets enable more effectively optimizing for BETs' operational characteristics, taking advantage of pooling effects to mitigate their operational constraints.

EVRP optimization enables operational efficiency of BET fleets. Prospective BET fleet operations need to consider that their operational characteristics causes operational decisions of BETs to diverge from that of ICETs in nearly 90% of all simulated scenarios with significantly different planning. This indicates that previous studies on electrification based on historical ICET operations potentially overestimate the influence of market prices due to mapping sub-optimal operations onto BETs. Because the operational patterns of ICETs are not representative of BETs, strategic decisions should therefore not be made solely based on current fleet operations. Likewise, from an operational perspective, 1-to-1 replacement of ICETs with BETs may lead to significant inefficiencies, and fleet operators should thus consider how BET operations can be optimized at the fleet level, such as charging pre-conditioning, shift allocation, and synchronizing driver breaks with charging. EVRP optimization can therefore be an effective tool for prospective screening of electrification.

Partial electrification is a robust strategy for competitiveness. By considering strategic–operational interfaces, our study shows that partial electrification of up to 50% is a robust strategy across a wide range of scenarios, even when the choice of battery sizing is decoupled from knowing the range requirements beforehand. However, predictable networks and driving distances offer more favorable conditions for electrification. When the battery can be sized appropriately, electrification rates up to 80% are most likely to be competitive at even longer distances.

Our analytical and modeling framework supports firms to make informed decisions about electrification under uncertain conditions, and thus accelerate the transition to a more sustainable transportation system.

One limitation of the study was only considering homogeneous fleets, and therefore excludes the operations of mixed BET and ICET fleets or different vehicle types and classes, which is of particular interest due to partial electrification being identified as a competitive option. Another aspect that might be overlooked in our study is optimizing routes according to price fluctuations throughout the day, which might lead to further fuel cost savings for BETs. Large portions of the variance in TCO could not be explained by the parameters included, but are likely caused by topological characteristics of networks beyond radius and size. Certain logistics service contents might have corresponding characteristics for their shipments that are more or less favorable for BETs. Additionally, other factors could be parameterized to study their effects on TCO, such as weather conditions which affect energy consumption and range. Exact placement of the enroute charging station could further optimize BET operations. The abovementioned variables should be studied thoroughly, but would apparently increase the modeling complexity. We leave such interesting concerns for our future studies.

CRedit authorship contribution statement

Anton Zackrisson: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Albin Engholm:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Ou Tang:** Writing – review & editing, Supervision, Conceptualization.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.trd.2024.104524>.

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